Experiment 4 — Process-Aware Rewards and Reward Hacking

A Validated Replication of §6 (Figs. 7–8) of Zhang, Neubig & Yue (2025)

DLT Final Project — Experiment 4

PACE ICE, 1× H100 (job 6589709), 1 h 20 m, 640,000 sampled solutions

October 2026

Abstract

We replicate §6 of *On the Interplay of Pre-Training, Mid-Training, and RL on Reasoning Language Models* (arXiv 2512.07783, ICML 2026 Spotlight) on the authors' released checkpoints: one shared 100M-parameter base model and four GRPO runs differing *only* in reward composition (outcome-only, two mixes, and a strict process gate), evaluated with k=64 samples on 2,000 problems in two contexts (640k samples). The paper's central claim is confirmed, and decisively: the **reward-hacking rate** — the share of correct answers reached through invalid reasoning — falls monotonically with process-supervision strength on the long-tail context (52.3% → 45.8% → ≈42.3% → 40.5%), with *every* step of the ladder significant under paired bootstrap after Bonferroni correction (strict vs. outcome: −5.31 pp, [−6.37, −4.27]). Strict pass@1 gains from process rewards are decisive on edge-difficulty problems (+6.5 pp, [+2.8, +10.3]) and directional but individually unresolved on the hardest band. All numbers are reproduced from the raw samples and validated by four independent audits.

1 Question and systems

Question. Compared with an *outcome-only* reward (pay for the correct final answer), do *mixed* or *strict process* rewards (also pay for verified reasoning steps) (a) raise strict pass@k and (b) lower the reward-hacking rate?

Five systems. One 100M-parameter base model (pre-trained 99.9% `crazy_zootopia` at op 2–20 + 0.1% `teachers_in_school` at op 2–20; the paper's appendix A.6.1 "shared primitives" setting) and four GRPO runs from it on *identical* data (90% zoo / 10% teachers, op 2–20, 204,800 samples ≈ 200 steps), differing only in reward:

Label	Reward (verified from released configs)	Status
`base`	none (pre-training only)	—
`outcome`	$R = \mathbb{1}[\hat{a} = a^*]$ (pure answer match)	verified
`0.5/0.5`	$R = \mathbb{1}[\hat{a} = a^*] \cdot (0.5 + 0.5\, R_{\text{pv}})$	weights assumed
`0.2/0.8`	$R = \mathbb{1}[\hat{a} = a^*] \cdot (0.2 + 0.8\, R_{\text{pv}})$	verified
`strict`	$R = \mathbb{1}[\hat{a} = a^*] \cdot \mathbb{1}[R_{\text{pv}} = 1]$	verified

Here R_{pv} is the *process accuracy* — the fraction of gold reasoning-graph nodes the solution reproduces with correct parents and values. The released code applies an *answer gate* (reward zeroed when the

answer is wrong), so the mixes are conditional on a correct answer; this diverges from the paper's literal Eq. 3, which was never trained in the released family.

2 Method

Choice	Value
Eval data	`crazy_zootopia` + `teachers_in_school` val, op $\in \{12, 14, 16, 18, 20\}$, 200/op
Samples	$k = 64$/problem, temperature 0.7, top-p 1.0, ≤ 1024 new tokens
Budget	$5 \times 2 \times 1{,}000 \times 64 = 640{,}000$ samples
Scoring	authors' graph pipeline: strict (process-verified) + answer-only pass@k
Bands	edge = op 12–14, hard = op 16–20
Uncertainty	bootstrap over problems (B=1000); **paired** bootstrap for contrasts (B=5000); Bonferroni α=0.05/9

Metrics. With first-success rank r_i of problem i, pass@$k = \frac{1}{N}\sum_i \mathbb{1}[r_i < k]$. With a_i answer-correct and s_i strictly-correct samples (of 64),

$$H \;=\; \frac{\sum_i \max(0, a_i - s_i)}{\sum_i a_i} \;=\; \Pr\big(\text{invalid reasoning} \mid \text{correct answer}\big).$$

3 Results

3.1 Headline — both contexts, overall

Table 1: Strict and answer-only pass@k, and hacking rate (%), overall. Hacking CIs are 95% bootstrap over problems.

Context	Reward	strict @1	strict @64	ans @1	ans @64	hacking rate
5*teachers	base	20.3	62.7	42.7	95.2	**52.3** [49.1, 55.7]
	outcome	36.1	68.8	66.9	94.4	**45.8** [42.7, 48.8]
	0.5/0.5	40.0	70.3	69.5	93.9	**42.3** [39.3, 45.7]
	0.2/0.8	39.2	68.0	68.6	93.4	**42.2** [38.7, 45.4]
	strict	38.7	68.8	64.4	94.4	**40.5** [37.3, 43.5]
5*zoo	base	60.3	87.0	89.5	99.6	33.6 [30.9, 36.6]
	outcome	64.3	85.1	92.8	98.7	32.6 [29.8, 35.3]
	0.5/0.5	62.3	86.4	92.7	98.7	32.3 [29.6, 34.9]
	0.2/0.8	62.6	86.2	93.0	98.6	32.2 [29.5, 34.9]
	strict	63.0	85.6	92.5	98.9	32.0 [29.2, 34.8]

3.2 The reward-hacking ladder (teachers) — the strongest result

Hacking falls monotonically with process-supervision strength. Because all five checkpoints answered *identical* problems, the right test is the paired bootstrap; **every step of the ladder is significant and survives Bonferroni** (†).

Paired contrast (teachers, overall)	Δ hacking (pp)	95% CI
base − outcome	+6.51	[+4.74, +8.25]†
0.5/0.5 − outcome	−3.48	[−4.35, −2.62]†
0.2/0.8 − outcome	−3.59	[−4.46, −2.73]†
strict − outcome	**−5.31**	[−6.37, −4.27]†

Context	Band / metric	base	outcome	0.5/0.5	0.2/0.8	strict
teachers	edge pass@1	33.5	44.8	**51.2**	47.8	48.0
teachers	edge pass@64	80.8	81.0	81.5	80.5	**83.0**
teachers	hard pass@1	11.5	30.3	32.5	**33.5**	32.5
teachers	hard pass@64	50.7	60.7	**62.8**	59.7	59.3
zoo	edge pass@1	68.8	71.0	67.8	68.2	69.8
zoo	edge pass@64	**94.5**	89.8	92.8	92.2	93.2
zoo	hard pass@1	54.7	59.8	58.7	58.8	58.5
zoo	hard pass@64	82.0	82.0	**82.2**	82.2	80.5

3.3 Strict pass@k by difficulty band

Key paired contrasts:

- **RL vs. base (teachers):** hard pass@1 +18.8 pp [+15.2, +22.5]; hard pass@64 +10.0 pp [+6.7, +13.3].
- **Process vs. outcome, edge:** 0.5/0.5 − outcome = +6.50 pp [+2.75, +10.25] (Bonferroni-robust).
- **Process vs. outcome, hard:** all three ahead (+2.2 to +3.2 pp), sign-consistent across all 5 ops (sign test $p = 0.031$), but none individually excludes zero (best $p = 0.056$) — directional support.
- **Zoo edge** pass@64**:** base − outcome = +4.75 pp [+2.25, +7.25] — RL narrows the sampled solution set where the base was already strong.

3.4 Per-op strict pass@1 (teachers)

Reward	op 12	op 14	op 16	op 18	op 20
base	43.5	23.5	12.0	14.0	8.5
outcome	51.5	38.0	34.5	30.5	26.0
0.5/0.5	57.0	45.5	34.5	36.5	26.5
0.2/0.8	52.0	43.5	35.5	37.0	28.0
strict	53.5	42.5	37.5	33.0	27.0

Process variants beat outcome-only in 14 of 15 (op × variant) cells and tie once — the sign-consistency behind the hard-band sign test.

3.5 Error breakdown (teachers, share of all samples %)

Process rewards shift errors away from structural failures, the dense mixes most clearly. *Nuance:* strict gating *increases* missing-step errors on the hard band (34.7% vs. outcome's 30.1%) while dense mixes decrease them (→ 24.8%) — an all-or-nothing trace reward favors short, conservative derivations; a dense per-step reward pays for completing each step.

Table 2: Hard band (op 16–20).

Reward	correct	wrong dep.	missing steps	wrong value	wrong answer
base	11.7	86.5	46.6	79.3	69.2
outcome	29.5	68.1	30.1	47.8	40.5
0.5/0.5	32.0	66.7	**24.8**	47.8	40.0
0.2/0.8	32.0	66.8	27.6	48.2	40.5
strict	30.9	67.9	*34.7*	52.0	44.5

Table 3: Edge band (op 12–14).

Reward	correct	wrong dep.	missing steps	wrong value	wrong answer
base	32.3	63.8	20.4	48.4	41.7
outcome	46.3	50.8	16.1	27.7	22.3
0.5/0.5	49.6	48.3	**11.9**	26.5	20.7
0.2/0.8	48.9	49.2	13.5	27.2	21.4
strict	49.2	48.8	14.6	27.9	22.6

4 Verdicts vs. the paper's predictions

1. *"Process signal raises strict* pass@1 *by ∼4–5 pp on the hardest problems."* — **Directional** on hard (+2.2 to +3.2 pp, sign-consistent, individually unresolved); **unambiguous on edge** (+6.5 pp [+2.8, +10.3]).
2. *"0.2/0.8 and strict best; strict lowest hacking."* — **Confirmed**; the hacking ladder is Bonferroni-robust at every step.
3. *"Errors shift away from wrong dependencies and missing steps."* — **Supported**, with the strict-gating missing-step reversal as a refinement.

5 Validation and limitations

Four audits: data integrity (all 64,000-record runs complete, byte-identical problems across checkpoints, zero truncation/duplicates); clean-room recomputation (independent code matching these numbers to $\sim 10^{-16}$ and the authors' aggregator exactly); qualitative re-scoring (240/240); paired statistics (CIs ∼3× tighter than independent via shared-difficulty cancellation).

Limitations. One training run per reward (CIs cover evaluation uncertainty only); context-setting substitution (direction-comparable to the paper's extrapolation-setting §6, not number-comparable); k=64 not 128; the 0.5/0.5 weights assumed. A letter-collision artifact in the authors' strict scorer inflates *absolute* hacking rates and deflates *absolute* strict pass@k for all checkpoints equally; the *paired differences and orderings* reported here are robust, but absolute levels should not be compared literally against the paper.

6 Reproducibility

One command on PACE: `sbatch scripts/run_all_pace.sbatch` (pinned: vLLM 0.10.1.1, transformers 4.55.2, torch 2.7.1). Pipeline: HF downloads → smoke test → vLLM generation → graph scoring → analysis (`src/analyze_e4.py`, bootstrap seed 0; paired bootstrap seed 12345) → figures.

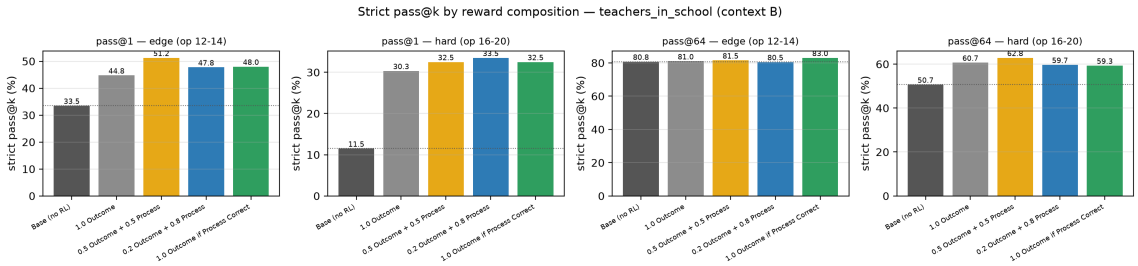


Figure 1: Strict pass@1/pass@64 by reward composition and band, teachers context.

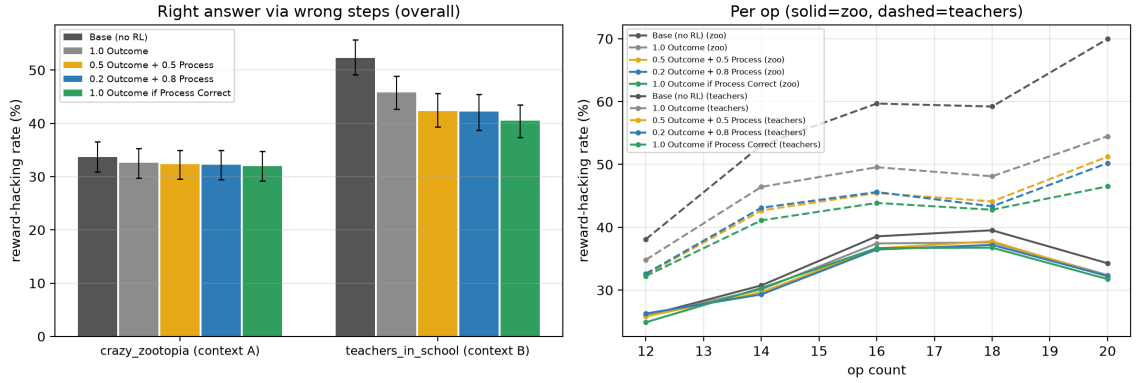


Figure 2: Reward-hacking rate by reward composition: overall per context (left, with 95% CIs) and per op count (right).

Raw samples in `results/`; metrics, CSVs, figures, and the four-audit validation in `analysis/`. Total compute: 1 h 20 m on one H100.

A Additional figures

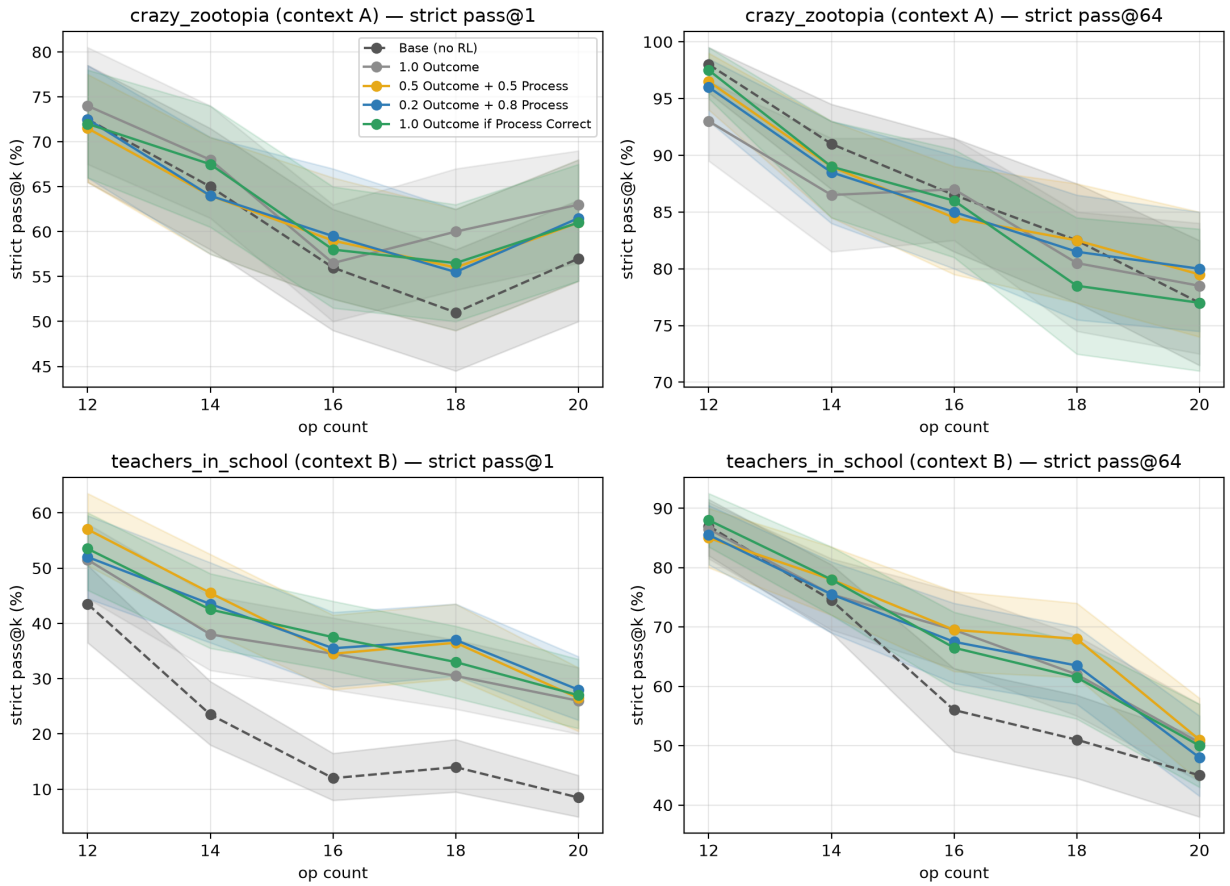


Figure 3: Strict pass@1 and pass@64 per op count, both contexts, with 95% bootstrap bands — the full data behind the headline and per-op tables. The teachers panels show RL lifting both pass@1 and pass@64 far above base (new ability on the seeded long-tail context); the zoo panels show RL raising pass@1 but not pass@64 (sharpening on already-covered material).

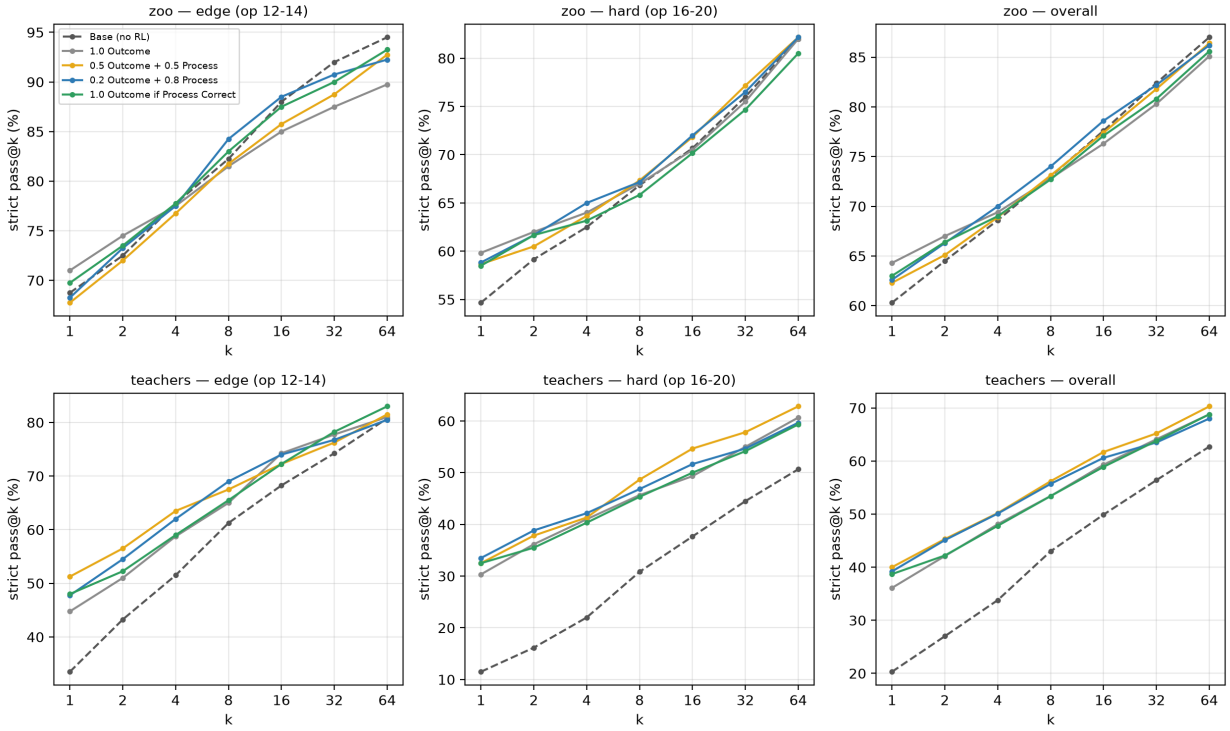


Figure 4: Strict pass@k vs. k ($k = 1 \ldots 64$, log scale) per difficulty band and overall, one line per reward. The gap between the base line and the RL lines on teachers, and its near-closure on zoo at large k, visualize the sharpening-vs-new-ability distinction.

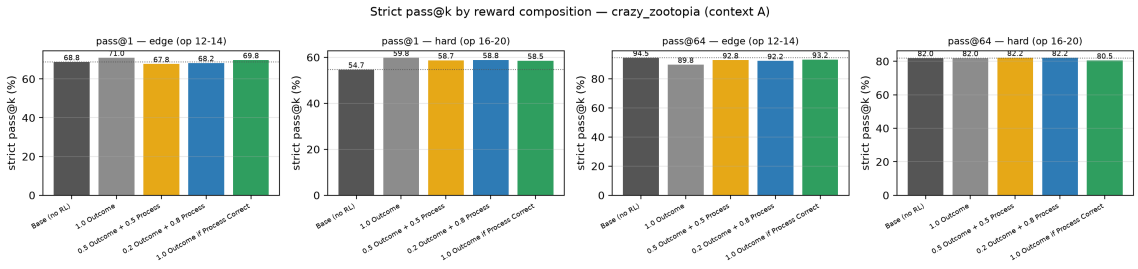


Figure 5: Strict pass@1/pass@64 by reward composition and band, zoo context (the well-covered context A). Contrast with the teachers bars in the main text: on zoo, reward design barely moves the metrics — process supervision matters most where the model is weakest.

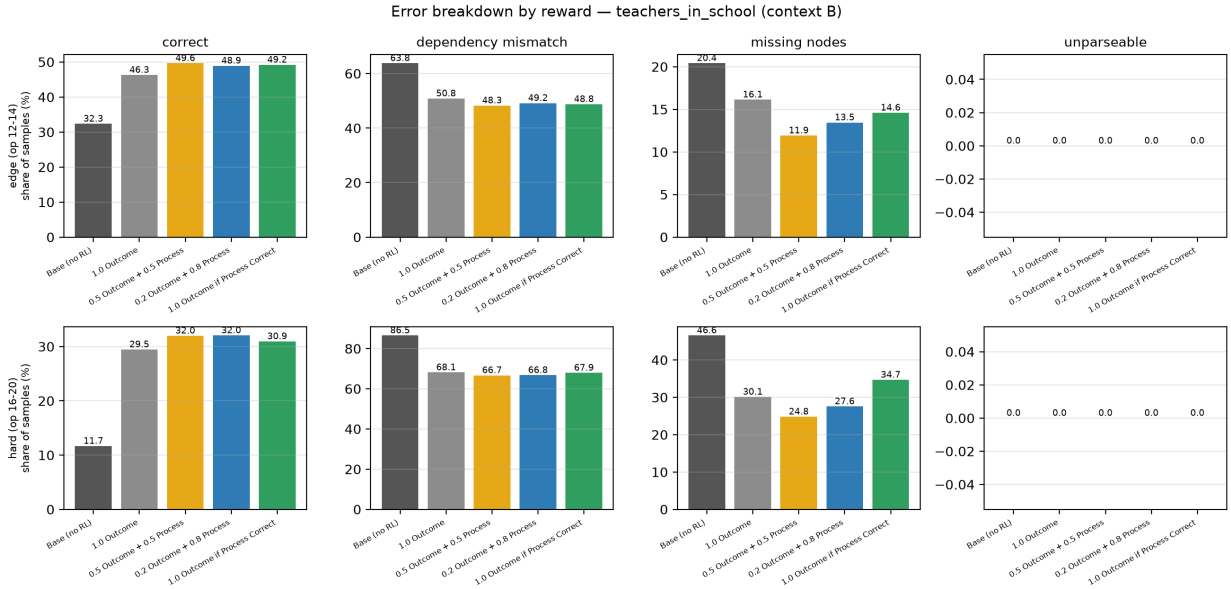


Figure 6: Error-category shares by reward and band, teachers context (paper-Fig.-8 analogue). Process rewards shrink the wrong-dependency and missing-step shares; note the strict-gating missing-step reversal on the hard band discussed in §3.5.

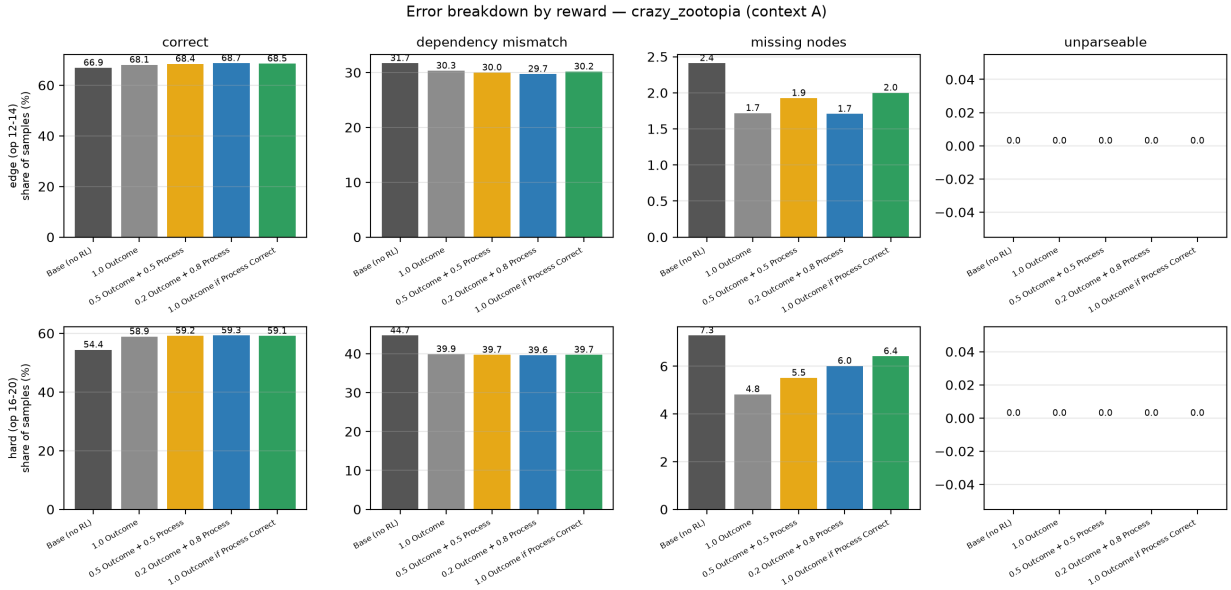


Figure 7: Error-category shares by reward and band, zoo context — shown for completeness; the categories are largely saturated on this easy-for-the-model context.